When Is a User's Own History Enough? A Controlled Study of Peer-Isolated Personal Recommendation

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Abstract. Personal data analytics systems often cannot consult other users: the data may be on-device, legally siloed, or simply absent for a new deployment. We ask when recommendation built *exclusively* from a single user's own interaction history is sufficient, and when it is not. We use Essence—a deliberately minimal peer-isolated system pairing a frozen sentence encoder with per-user K-means and cosine retrieval—as an instrument rather than a proposal, evaluating it against nine baselines on three datasets spanning music, e-commerce and film. Two findings emerge. First, the answer is governed by the density of cross-user co-occurrence in the evaluated population, not by the content domain. Peer-isolated retrieval ranks second and third of ten where that density is low (Last.fm-1K, Amazon Books) and eighth of ten where it is high (MovieLens-25M); the pattern belongs to the whole no-pooling architecture family rather than to any particular component. We demonstrate the mechanism *within a single dataset*: raising Last.fm-1K's user cap lifts the share of items heard by two or more users from 10.3% to 28.4% and reverses the verdict, from $d = +0.28$ in favour of peer-isolated retrieval over item-based collaborative filtering to $d = -0.16$ against it, both significant. A domain is therefore not weak or strong in collaborative signal; a *cohort* is. Second, and centrally, the interest-clustering step contributes nothing measurable. In the targeted test on the users whose taste is most clearly multi-modal it is significant in none of 18 silhouette-stratified comparisons; and across the fourteen dataset-level comparisons spanning three datasets, two sentence encoders, two user-activity cohorts and two temporal-split protocols, exactly one favours clustering and one favours simple recency-weighting, both marginal. We report the single exception rather than suppress it, and show it falls below its own detection threshold. We also report a two-sided bootstrap defect, present in our earlier work, that counted tied resamples as evidence against the null and inflated significance on sparse long-tail metrics. Baselines throughout are given the benefit of the doubt, a convention that costs us one previously significant result.

Keywords: Personal data analytics · Recommender systems · Negative results · Cold start · Reproducibility

1 Introduction

A recommender that may not look at other users is an unusual object. Almost all deployed systems pool behaviour across a population, and the dominant methods—collaborative filtering, matrix factorisation, sequential models—are defined by that pooling. Yet a growing class of settings forbids it. On-device assistants hold data that never leaves the phone. Regulated domains silo records by subject. A new product has no population to pool over. In each case the practical question is the same: *how much is lost by modelling a user from their own history alone?*

This paper answers that question empirically and reports a negative result at its centre. We state the genre plainly, because it governs how the paper should be read: this is a measurement study, not an architecture proposal. A reader looking for a new mechanism will not find one; what is on offer is a mechanism-level account of *when* peer-isolated retrieval is viable, a null bounded by its own detection threshold, and a protocol for testing both.[1]

We build a minimal peer-isolated system, Essence, and treat it as an instrument. Essence embeds items with a frozen sentence encoder, partitions each user's history into K interest clusters with K-means, selects the cluster nearest the user's recent activity, and retrieves by cosine similarity. No component is novel and none is intended to be; the system exists to isolate the variable of interest, which is the absence of cross-user information.

Contributions.

1. **A population-density result.** Peer-isolated retrieval is competitive where cross-user co-occurrence is sparse and far behind where it is dense. We identify the mediator by manipulating it *within* a dataset — twice, by two different means — rather than inferring it from a three-dataset contrast, and show the effect belongs to the no-pooling family rather than to clustering.
2. **A clustering null, established at scope.** Per-user interest clustering gives no measurable benefit over recency-weighted retrieval across three datasets, two encoders, two activity cohorts and two split protocols, with a minimum detectable effect for every comparison so the claim is bounded rather than asserted.
3. **A statistical correction.** We fix a tie-handling defect in the two-sided bootstrap p-value used in our earlier work, which reported some zero-versus-zero comparisons as maximally significant.

[1] An earlier, substantially weaker version appeared as a non-archival poster at the New In ML workshop, NeurIPS 2026; "our earlier work" throughout denotes that version. This paper supersedes it — the K-selection rule, significance testing, baseline configurations and headline finding have all changed, and we document each change and its direction. The encoder, heavy-user and expanded-cohort replications, power analysis, sequential baselines, multiplicity correction, seed-stability check and cohort-density result are new. Code, data-preparation scripts and all result tables: https://github.com/biditdas18/essence (tag `soict-2026`).

4. **Conservative reporting.** Baselines are reported at the better of their published and validation-tuned configurations, costing us one previously significant result.

2 Related Work

Multi-interest and sequential recommendation. MIND [1] and ComiRec [2] represent a user by several interest vectors rather than one, on the premise that a single embedding cannot capture heterogeneous taste; GRU4Rec [3] and SASRec [4] model interaction order. All four learn from pooled data. Essence adopts the multi-interest premise but removes the pooling, which is what lets us ask whether the premise survives in isolation. Content-only retrieval, for its part, avoids cold start structurally by not requiring co-occurrence — a generic property of the approach rather than of any system, a distinction the literature does not always preserve.

Negative and reproducibility results. Several authors have found that carefully tuned simple baselines match or exceed neural recommenders [5,6]. Our clustering null is of that kind, with the difference that the component we find inert is our own.

3 Method

Let user u have interaction history $H_u = (i_1, \ldots, i_n)$ in chronological order. Each item i carries text t_i (title, artist or author, genre or category, description). We embed with a frozen sentence encoder ϕ, so $v_i = \phi(t_i)/\|\phi(t_i)\|$. No interaction data enters ϕ.

Interest clusters. We run K-means on $\{v_i : i \in H_u\}$ to obtain centroids $c_1, \ldots, c_K$ for that user alone. If $|H_u| < K$ the user cannot be clustered and falls back to the mean-embedding method; we report the frequency of this.

Active cluster and retrieval. Let r_u be the mean of the last ten items' embeddings. We select $c^\star = \arg\min_k \|c_k - r_u\|$ and return the ten unseen catalogue items of highest cosine similarity to $c^\star$.

Selecting K. Our earlier work selected K with a stopping rule evaluated forward from a checkpoint at $K = 8$, which a reviewer correctly identified as anchoring. We replace it with a criterion that is anchoring-free by construction: choose the K maximising validation Recall@10 subject to (a) mean items per cluster ≥ 3 and (b) every user clusterable. Constraint (a) prevents degenerate singleton clusters; (b) prevents a configuration whose apparent quality is partly produced by fallback. Table 1 gives the outcome.

The adopted values are invariant to the sentence encoder. They are not, however, a property of the domain: on Amazon Books the same criterion selects $K = 8$ for users with $\sim$29 interactions and $K = 60$ for users with 330 (Sect. 5.4). Any single reported K is contingent on the user filter.

Table 1. Adopted K under the anchoring-free criterion. Both changes are forced by constraints the original rule did not enforce.

Dataset	Published K	Adopted K	Reason for change
Last.fm-1K	10	10	unchanged
Amazon Books	10	**8**	$K = 10$ gives 2.58 items per cluster, below the floor of 3
MovieLens-25M	15	**10**	at $K = 15$ only ∼90% of users could cluster

4 Experimental Setup

Datasets. Three domains, chosen to span a range of collaborative signal strength: music (Last.fm-1K), book retail (Amazon Books) and film (MovieLens-25M). Users are filtered to an activity band and, for the two larger sets, sampled; this gives 99, 2,000 and 2,000 users. Each history is split chronologically at that user's own 80th percentile. We also report a global-cutoff variant (Sect. 5.7).

Baselines. Nine: Recency-Weighted, Last-Item, Avg-Last-10, Content (mean embedding), CF-ItemKNN, MIND, ComiRec, Popularity and Random, plus SASRec and GRU4Rec as library-implemented sequential models (Sect. 5.6). Recency-Weighted is the critical one: it is Essence with the clustering step deleted, so the Essence–Recency-Weighted contrast isolates clustering exactly.

Conservative baseline convention. Each baseline is reported at the *better of its published and validation-tuned configuration on test*. This is deliberately not the textbook protocol. We adopt it because strict validation-selection would in one case have reported a ComiRec configuration 15.8% worse on test than one we had already run, turning a tuning artifact into an apparent win for our own system. The convention favours the baselines and can only weaken our results. It costs us one previously significant comparison, reported in Sect. 5.1.

Statistics. Paired bootstrap over users, 10,000 resamples, two-sided p counting exact ties in both tails, Benjamini–Hochberg FDR within family, Cohen's d for paired differences. For every null we also report the minimum detectable effect at 80% power, estimated from the bootstrap standard error (Sect. 5.3).

5 Results

5.1 Domain dependence

Table 2 shows the pattern: Essence ranks second and third of ten on Last.fm-1K and Amazon Books, and eighth of ten on MovieLens-25M, where CF-ItemKNN alone beats it by more than an order of magnitude. Those ranks are unchanged under Recall@5, Recall@20 and NDCG@10 on the two datasets where the full

Table 2. Recall@10 at adopted K. Baselines at the better of published and tuned. Best per dataset in bold.

System	Last.fm-1K	Amazon Books	MovieLens-25M
Essence	0.0281	0.0246	0.0055
Recency-Weighted	**0.0304**	0.0260	0.0076
Last-Item	0.0265	**0.0311**	0.0091
Avg-Last-10	0.0252	0.0235	0.0064
Content (Avg Emb)	0.0038	0.0173	0.0043
CF (ItemKNN)	0.0018	0.0031	**0.0850**
MIND	0.0082	0.0072	0.0423
ComiRec (SA)	0.0243	0.0056	0.0706
Popularity	0.0023	0.0021	0.0517
Random	0.0001	0.0001	0.0012

metric set was available, with MIND and ComiRec re-scored from the same ranked lists, so the ordering depends on neither the cutoff nor the use of a ranking-aware measure.

The effect is not a property of clustering. Recency-Weighted—identical to Essence but with no clustering at all—exhibits the same profile, ranking first on Last.fm-1K and collapsing on MovieLens-25M. What the data separate is architectures that pool across users from architectures that do not.

What the conservative convention costs. Essence significantly outperforms MIND on both sparse datasets and ComiRec on Amazon Books, but is *indistinguishable from a properly tuned ComiRec on Last.fm-1K* ($d = 0.065$, $q = 0.54$): tuning closed that gap from 2.05× to 1.15×, and our earlier claim rested on a ComiRec denied a fair learning rate. The comparison is underpowered too (detectable effect 0.0159 against an observed 0.0038), so the two are *not shown to differ* at $n = 99$—not equal. The defensible claim is narrower than before: peer-isolated retrieval is *competitive with* tuned multi-interest recommenders where co-occurrence is sparse, and far behind where it is dense.

5.2 The clustering null

The sharpest version of the test restricts attention to the users whose taste is most clearly multi-modal, stratifying by silhouette score: if clustering helps anyone, it should help them. At validation-selected K, with tie-aware p and no rows excluded, clustering is significant in none of those 18 stratified comparisons against recency-weighted retrieval. Our earlier work reported "one of sixteen", where the sixteen were a hand-curated subset—formed, we later found, by dropping exactly the degenerate cases the defect in Sect. 5.13 produced—and the one significant row was itself an artifact of that defect.

Table 3 states the null at full scope.

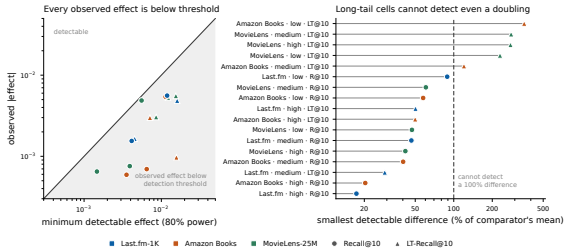


Fig. 1. Observed effect against minimum detectable effect at 80% power for every null comparison. Points below the diagonal are effects the test could not have detected. Ranking-metric nulls are adequately powered; several long-tail cells are not and are not claimed.

Table 3. Every Essence versus Recency-Weighted comparison in this study.

Robustness check	Configuration	Null holds?
Main analysis	3 datasets, adopted K	yes
Second sentence encoder	all-mpnet-base-v2, 2 datasets	yes (0 of 8 favour clustering)
Heavy users	330 interactions/user, $K = 60$	yes ($q = 0.31$, powered to 21.6%)
Global temporal cutoff	single wall-clock split	Recall@10 yes; **one exception**
Silhouette-stratified	most multi-modal users	yes

Across all fourteen comparisons—three datasets, two encoders, two activity cohorts, two split protocols—exactly one favours clustering and one favours recency-weighting, both marginal.

The exception, reported and not claimed. Under the global cutoff, clustering beats recency-weighting on long-tail recall at $q = 0.0496$. It is not an artifact of fallback users: restricted to the 357 users who genuinely cluster, it survives at $p = 0.0348$, $d = 0.096$. We nonetheless decline to claim it. The observed effect is 0.64× its own 80%-power detection threshold, placing it in the regime where significant findings are most inflated; it is one comparison out of fourteen, where 0.7 false positives are expected at $\alpha = 0.05$; and $d = 0.09$ is negligible beside the effects the rest of the paper turns on.

5.3 Statistical power

A null is only informative if the test could have detected an effect. Of 18 null comparisons, 17 are informative and none reach their own minimum detectable effect (Fig. 1), splitting into two regimes: nulls on ranking metrics are adequately powered and carry content, while several long-tail cells cannot detect even a doubling of the comparator and we do not assert those as nulls — a correction to our earlier work, which did.

5.4 Heavy users

All three datasets bound user activity from above, and the excluded heavy users are the most plausibly multi-modal — so the null may have been established on the population least likely to exhibit the effect it tests. We replicated on Amazon Books users with 201–1000 interactions (1,757 users, 330 interactions each against ∼29, catalogue 213,221 items), re-selecting $K = 60$ so clustering received its best configuration. The null holds: Recall@10 difference −0.000288 ($d = -0.027$, $q = 0.31$), powered to detect a 21.6% relative difference. Clustering does not help the users it was supposed to help.

Essence slips from third to fourth of eight comparable systems, and CF-ItemKNN degrades by only 22% where content methods lose 86–93%: heavy users supply the co-occurrence signal CF needs. This is the mechanism of Sect. 5.9 a second time, in another dataset and under the opposite manipulation — raising the activity floor rather than lifting the ceiling.

5.5 Encoder sensitivity

Replacing all-MiniLM-L6-v2 (384-dim) with all-mpnet-base-v2 (768-dim) leaves conclusions intact: Essence improves slightly (Amazon Books 0.0246 to 0.0254, rank three to two), one of eighteen verdicts changes, and selected K is identical. As a correctness check, the five systems not consuming item embeddings return bit-identical results while exactly the five that do move.

5.6 Sequential baselines

SASRec and GRU4Rec were run through RecBole with learning rates tuned on held-out validation, and scored with our own metric implementations from the library's ranked lists. Both fail on these datasets: Essence leads by 20.9× and 23.3× on Last.fm-1K and by 174.6× and 59.3× on Amazon Books. Even taking each model's best test configuration—a deliberately unfair comparison that favours the baselines—the gap is 11.9× to 43.6×.

The cause is structural rather than a tuning failure. Amazon Books is 0.063% dense with 32 training items per user, so there is very little transition structure to learn. This is the same weak-collaborative-signal condition our thesis concerns, and the result is therefore consistent with it rather than incidental. It is emphatically *not* evidence that Essence outperforms SASRec in general; on denser data we would expect the ordering to reverse, as it already does for the other collaborative baselines on MovieLens-25M.

5.7 Temporal leakage

Per-user chronological splitting allows one user's training window to overlap another's test window, which can favour collaborative methods. Under a single global cutoff every system degrades, but unevenly: CF-ItemKNN loses 86% where content and recency methods lose 37–45%, and Essence's rank improves from third to second of eight. The leakage was inflating the collaborative baselines, so our main protocol is the conservative choice for this paper's central comparison.

Table 4. The between-dataset difference as an interaction, not a ranking. Δd is the change in Essence's effect size when moving from a sparse-co-occurrence dataset to MovieLens-25M. Every contrast is significant after BH correction and every one is a sign flip: Essence ahead becomes Essence behind. These hold for the datasets *as configured*; Sect. 5.9 shows Last.fm-1K's position is set by its user cap, not by its domain.

Baseline	Sparse-density set	d (weak)	d (MovieLens)	Δd	95% CI
CF (ItemKNN)	Last.fm-1K	+0.28	−0.55	0.83	[0.76, 0.95]
CF (ItemKNN)	Amazon Books	+0.25	−0.55	0.80	[0.76, 0.84]
MIND	Last.fm-1K	+0.24	−0.38	0.62	[0.54, 0.76]
MIND	Amazon Books	+0.21	−0.38	0.58	[0.54, 0.63]
ComiRec (SA)	Last.fm-1K	+0.07	−0.52	0.58	[0.33, 0.72]
ComiRec (SA)	Amazon Books	+0.22	−0.52	0.74	[0.70, 0.78]
Popularity	Last.fm-1K	+0.27	−0.44	0.71	[0.63, 0.82]
Popularity	Amazon Books	+0.26	−0.44	0.69	[0.65, 0.74]

5.8 Is the domain difference real? A formal test

Ranks describe the pattern; they do not test it. We therefore test the interaction directly. For each collaborative baseline we compute the paired effect size (Cohen's d) of Essence against it within each dataset, bootstrap each dataset's d independently (10,000 resamples), and take the distribution of the difference. d is scale-free, which matters because absolute recall differs by an order of magnitude across these datasets.

All 8 contrasts are significant after BH correction, all 8 show a sign flip, and no confidence interval contains zero; the smallest lower bound is 0.33. Δd ranges from 0.58 to 0.83, which are large effects. The difference between datasets is therefore a tested claim rather than a described pattern. It holds for these datasets *as configured*: Sect. 5.9 shows the position on this axis is set by the cohort, not the domain.

5.9 Cohort density, not domain, drives the verdict

Sect. 5.8 establishes that Essence's standing differs between datasets. It does not establish *why*, and the natural reading — that music and book retail simply carry weaker collaborative signal than film — turns out to be wrong. The Last.fm-1K cohort is capped at 500 distinct tracks per user, which admits 99 of the 975 users clearing the minimum. Lifting that cap to 2,000 admits 374, and reverses the result.

Both verdicts are significant ($q < 0.0001$ and $q = 0.014$). On the expanded cohort CF-ItemKNN is the best of the eight systems and Essence falls from second to fifth. The mediating quantity is in the fifth column: item-based collaborative filtering requires cross-user co-occurrence, and the expanded cohort supplies nearly three times as much of it.

Table 5. The same dataset, the same domain, the same code and seed. Raising the user cap raises cross-user co-occurrence and reverses the verdict against item-based collaborative filtering. Mean training history rises from 215 to 783 items per user and the candidate catalogue from 18,679 to 158,074 items, so absolute recall falls for every system; only within-row comparisons are meaningful.

Cohort (tracks/user)	Users	Items w/ ≥ 2 users	Essence	CF-ItemKNN	d
50–500 (published)	99	**10.3%**	0.0281	0.0018	**+0.28**
50–2,000	374	**28.4%**	0.0026	0.0055	**−0.16**

This reframes the first finding rather than overturning it. Signal strength is a property of the *evaluated population*, not the content domain, and the three-dataset pattern is three points on that one axis — which is why MovieLens-25M, the densest, is where peer-isolated retrieval fails worst. It is also actionable: measure co-occurrence density in your own user base rather than reason from your product category.

A confound we cannot remove. Lifting the cap raises user count (99 to 374) and history length (215 to 783 items/user) together. Both act through the same mediator and this design cannot separate them, since isolating user count would need 374 users with 50–500 interactions and only 99 exist. We claim density as the mediator, not which of its causes dominates. MIND and ComiRec are omitted from Table 5: trained on the 99-user sample, they cover only 26.5% of the expanded cohort, and mixing a 99-user evaluation into a 374-user ranking is the error this paper elsewhere reports catching. The clustering null is unaffected and moves marginally further against clustering (LT-Recall@10 $q = 0.031$).

5.10 Study-wide multiplicity

FDR above is controlled within each family, which is conventional but invites a fair question: how many hypotheses does this study test in total? The answer is 136. Applying Benjamini–Hochberg across all 136 at once takes the significant count from 59 to 56 and changes exactly two rows, both against clustering.

First, the one exception in Sect. 5.7 — the single comparison in this study where clustering beat recency-weighting — is not significant study-wide ($q = 0.075$). We had already declined to claim it on power grounds; the stricter correction independently agrees. Under it, *no comparison anywhere in this study shows clustering significantly outperforming recency-weighted retrieval.*

Second, one comparison becomes significant that was not: in the silhouette-stratified test the *high*-silhouette stratum — the users whose histories separate most cleanly into distinct interest groups, which is precisely the population the multi-interest premise concerns — shows clustering performing significantly *worse* than recency-weighting ($q = 0.033$). This is not a paradox: BH adapts to the distribution of p-values it is given, and a family of 18 mostly-null tests is a less favourable reference than the full study. It is the strongest form the

negative result takes. On the users where interest clustering should help most, it measurably hurts.

We report within-family FDR as the primary correction and state the study-wide result alongside it. Reporting only the former would preserve an exception we do not believe; reporting only the latter would overstate the certainty of a post-hoc choice.

5.11 Seed stability

Two seeds were fixed and never varied: the per-user K-means initialisation and the random user sample. Re-running both on Amazon Books at the adopted K, the clustering null holds in all five configurations with identical significance counts; no verdict depends on either seed. Worth stating since the opposite would be a fair criticism: the published seed is mildly *unfavourable* to Essence — against Recency-Weighted it gives $q = 0.52$ where the alternatives give 0.61 and 0.90, further from significance, with Essence scoring 8.0% higher under them.

5.12 The identity shortcut, at the adopted K

Our earlier work reported that roughly 20% of Essence's hits share an artist with the user's active cluster, and that excluding same-artist candidates collapses performance by 85–90%. That analysis was run at $K = 3$, a value this paper does not adopt — an inconsistency we found while re-verifying it. At $K = 3$ the figures reproduce exactly (20.6% shortcut share, 87.9% collapse). At the adopted $K = 10$ the dependence is substantially larger: **32.9% of hits are same-artist and removing those candidates costs 95.3% of performance**.

More clusters means tighter, more artist-homogeneous clusters, so nearest-centroid retrieval lands closer to artists already inside the active cluster. The shortcut is a property of the retrieval geometry and it grows with K. Restating it at the adopted K therefore sharpens rather than softens the limitation: a large part of what this architecture recovers is identity matching, not taste modelling.

Cold start. At the adopted K, content-only retrieval beats CF-ItemKNN in 5 of 6 popularity-decile comparisons at the two coldest deciles, the exception being MovieLens's second decile where CF has begun to acquire signal — the mechanism of Sect. 5.9 at the item level, and generic to content-only retrieval rather than specific to Essence.

5.13 A statistical defect in our earlier work

The two-sided bootstrap p-value was computed as $\min(1, 2\min(f, 1-f))$ with $f = \text{mean}(\delta > 0)$ over resampled differences δ. This counts every resample with $\delta = 0$ as evidence against the null. On long-tail recall, where most users score exactly zero, the resample distribution has large mass at zero, so a one-sided-with-ties difference was reported as $p = 0$—including cases in which both arms were identically zero for every user and no difference existed to detect.

We fixed this in all call sites by counting ties in both tails. The correction is consequential and moves against us: the clustering-value test falls from seven of eighteen significant to none of 18. We verified the fix is surgical rather than wholesale—one dataset's corrected output is byte-identical to its published counterpart.

6 Discussion and Conclusion

The practical question has a conditional answer: a user's own history is enough when cross-user co-occurrence is thin and not enough when it is dense. That boundary belongs to the *cohort*, not the domain — the same music data sits on either side of it depending on the user cap — so the diagnostic for a practitioner under an on-device or siloed-data constraint is not which algorithm to pick, but whether their user base supplies dense co-occurrence. Within that design, per-user interest clustering — the component we originally proposed as the contribution — adds nothing measurable over simple recency weighting across every robustness dimension we tested. The multi-interest premise itself may well hold; our data show only that recovering it by unsupervised partition of one user's embedded history yields nothing a recency-weighted query does not already have.

Limitations. Item text quality bounds every content method here and we did not vary it. The long-tail nulls are underpowered and we do not claim them. The main Last.fm-1K analysis has 99 users, limiting power there; Sect. 5.9 reports the expanded cohort. Sequential baselines on Amazon Books use a split agreeing with ours on 92.8% of test items rather than exactly.

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